Carboxylic Acids

Exam code: 0620 & 0971

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Carboxylic acids

- The carboxylic acids behave like other acids
- They react with:
  - **metals** to form a salt and hydrogen
  - **carbonates** to form a salt, water and carbon dioxide gas
  - **bases** to form a salt and water
- The salts formed by the reaction of carboxylic acids all end **–anoate**
- So methanoic acid forms a salt called methanoate, ethanoic a salt called ethanoate etc.
- In the reaction with metals, a metal salt and hydrogen gas are produced

Example reactions of carboxylic acids

- The reaction of ethanoic acid with **metals** such as magnesium forms the salt magnesium ethanoate and hydrogen gas:

$$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$$

- The **neutralisation** reaction of a carboxylic acid with a **hydroxide** produces salt and water
- For example, the reaction between potassium hydroxide and ethanoic acid forms the salt potassium ethanoate and water:

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produces a metal salt, water and carbon dioxide

- For example, in the reaction between potassium carbonate and ethanoic acid, the salt potassium ethanoate is formed with water and carbon dioxide

$$2CH_3COOH + K_2CO_3 \rightarrow 2CH_3COOK + H_2O + CO_2$$

Examiner Tips and Tricks

You need to be able to name and give the formulae of the salts produced in these reactions.

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